

Package Insert:

Product name: PRO-XYLO (Xylazine injection USP 100mg/ml)

Name and Strength of Active Ingredients

Xylazine HCL	100 mg / ml
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Excipients:

Methyl Paraben	1mg/ml
Propyl Paraben	0.1 mg/ml
Sodium Metabisulphite	0.4mg/ml
Sodium citrate	0.4mg/ml
WFI	q.s

Product Description: A Clear, Colourless particle free solution

Pharmacodynamics:

- Xylazine belongs to the $\alpha 2$ -adrenoceptor agonists.
- Xylazine is a $\alpha 2$ -adrenoceptor agonist that acts by the stimulation of central and peripheral $\alpha 2$ -adrenoceptors. Through its central stimulation of $\alpha 2$ -adrenoceptors, Xylazine has potent antinociceptive activity. In addition to $\alpha 2$ -adrenergic activity, Xylazine has $\alpha 1$ -adrenergic effects.
- Xylazine also produces skeletal muscle relaxation by inhibition of intraneuronal transmission of impulses at the central level of the central nervous system. The analgesic and skeletal muscle relaxation properties of Xylazine show considerable interspecies variations. Sufficient analgesia generally will be attained in combination with other products only.
- In many species, administration of Xylazine produces a short-lived arterialpressor effect followed by a longer period of hypotension and bradycardia. These contrasting actions upon the arterial pressure apparently are related to the $\alpha 2$ - and $\alpha 1$ -adrenergic actions of Xylazine.
- Xylazine has several endocrine effects. Insulin (mediated by $\alpha 2$ -receptors in pancreatic β -cells which inhibit insulin release), ADH (decreased production of ADH, causing polyuria) and FSH (decreased) are reported to be influenced by Xylazine.

Pharmacokinetics: Xylazine is a highly lipid soluble organic base and diffuses extensively and rapidly. Absorption is rapid following intramuscular injection. Within minutes after an intravenous injection, it can be found in a high concentration in the kidneys, the liver, the CNS, the hypophyses, and the diaphragm. So there is a very rapid transfer from the blood vessels to the tissues. Xylazine is metabolised extensively and eliminated rapidly. The rapid elimination of Xylazine is probably related to an extensive metabolism rather than to a rapid renal excretion of unchanged Xylazine.

Indication: Sedation, Premedication in combination with an anaesthetic.

Recommended Dosage:

Cattle (≥ 200 kg): Intramuscular.

Horses: Intravenous.

To ensure a correct dosage body weight should be determined as accurately as possible. The intravenous injection to horses should be given slowly.

Cattle:

Dosage:

Dosage for Cattle			
Dosage level	Xylazine (mg/kg)	ProXylo (Xylazine injection USP) 100 mg/ml (ml/100 kg)	ProXylo (Xylazine injection USP) 100 mg/ml (ml/500 kg)
I	0.05	0.05	0.25
II	0.1	0.1	0.5
III	0.2	0.2	1
IV	0.3	0.3	1.5

Dose 1: Sedation, with a slight decrease of muscle tone. The ability to stand is maintained.

Dose 2: Sedation, marked decrease of muscle tone and some analgesia.
The animal usually remains standing, but may lie down.

Dose 3: Deep sedation, further decrease of muscle tone and a degree of analgesia. The animal lies down.

Dose 4: Very deep sedation, a profound decrease in muscle tone and a degree of analgesia. The animal lies down.

Horses

Dosage:

Single dose of 0.6-1 mg Xylazine per kg body weight.
(0.6-1 ml product per 100 kg body weight).

Route of Administration: Intramuscular (Cattle ≥ 200 kg) and Intravenous (Horses)

Contraindications:

- Do not use in case of hypersensitivity to the active substance or to any of the excipients.
- Do not use in animals with gastrointestinal obstruction as the muscle relaxant properties of the drug appears to accentuate the effects of the obstruction and be the cause of possible vomiting.

- Do not use in animals with severe renal or hepatic impairment, respiratory dysfunction, cardiac abnormalities, hypotension and/or shock.
- Do not use in diabetic animals.
- Do not use in animals with a history of seizures.
- Do not use in cattle weighing less than 200 kg bodyweight. Do not use in foals younger than 2 weeks.
- Do not use during the last stage of pregnancy (danger of premature birth), except at parturition.

Warnings and Precautions:

Horses:

- Xylazine inhibits the normal intestinal motility. Therefore, it should only be used in horses with colic, that are not responsive to analgesics. The use of Xylazine should be avoided in horses with caecal malfunction.
- After treatment of horses with Xylazine, the animals are reluctant to walk, so whenever possible the drug should be administered in the place where the treatment/investigation is going to take place.
- Caution should be taken in the administration of the product to horses susceptible to laminitis.
- Horses with airway disease or malfunction may develop life-threatening dyspnoea.
- The dose should be kept as low as possible.
- The association with other pre-anaesthetic agents or anaesthetic agents should be the subject of a benefit/risk assessment. This assessment should consider the composition of the products, their dose and the nature of the surgery. Recommended dosages are likely to vary according to the choice of the anaesthetic association.

Cattle (≥ 200 kg) :

- Ruminants are highly susceptible to the effects of Xylazine. Normally cattle remain standing at the lower doses, but some animals may lie down. At the highest recommended doses most animals will lie down and some animals may lapse in lateral recumbency.
- Reticulo-ruminal motor functions are depressed after injection of Xylazine. This may result in bloat. It is advisable to withhold feed and water for several hours before administration of Xylazine.
- In cattle the ability to eructate, cough and swallow is retained but reduced during the period of

sedation, therefore cattle must be closely watched during the recovery period: the animals should be maintained in sternal recumbency.

- In cattle life threatening effects may occur after intramuscular doses above 0.5 mg/kg body weight (respiratory and circulatory failure). Therefore very precise dosing is required.
- This product should only be used in cattle weighing 200 kg or more. Since it is highly concentrated, a slight deviation from the actual volume to be injected may cause serious adverse reactions.
- The association with other pre-anaesthetic agents or anaesthetic agents should be the subject of a benefit/risk assessment. This assessment should consider the composition of the products, their dose and the nature of the surgery. Recommended dosages are likely to vary according to the choice of the anaesthetic association.

Precautions:

Special precautions for use in animals

Keep the animals calm, because they may respond to external stimuli.

Avoid intra-arterial administration.

Tympany may occasionally occur in recumbent cattle and can be avoided by maintaining the animal in sternal recumbency.

To avoid aspiration of saliva or food, lower the animal's head and neck. Fast the animals before use of the product.

- Older and exhausted animals are more sensitive to Xylazine, whilst nervous or highly excitable animals may require a relatively high dose.
- In case of dehydration, Xylazine should be used cautiously.
- Do not exceed the recommended dosage.
- Following administration, animals should be allowed to rest quietly until the full effect has been reached.
- It is advised to cool animals when the ambient temperature is above 25°C and to keep animals warm at low temperatures.
- For painful procedures, Xylazine should always be used in combination with local or general anaesthesia.
- Xylazine produces a certain degree of ataxia; therefore, Xylazine must be used cautiously in procedures involving the distal extremities and in standing castrations in the horse.
- Treated animals should be monitored until the effect has faded totally (e.g. cardiac and respiratory function, also in the post-operative phase) and should be segregated to avoid bullying.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

- Care should be taken to avoid accidental self-injection. In case of accidental oral intake or self-injection, seek medical advice immediately and show the package leaflet to the physician but DO NOT DRIVE as sedation and changes in blood pressure may occur.
- Avoid skin, eye or mucosal contact.
- Wash the exposed skin immediately after exposure with large amounts of water.
- Remove contaminated clothes that are in direct contact with skin.
- In the case of accidental contact of the product with eyes, rinse abundantly with fresh water. If symptoms occur, seek the advice of a physician.
- If pregnant women handle the product, special caution should be observed not to self-inject as uterine contractions and decreased foetal blood pressure may occur after accidental systemic exposure.

Advice to doctors: As Xylazine being an α_2 -adrenoreceptor agonist; symptoms after absorption may involve clinical effects including dose-dependent Sedation, Respiratory depression, Bradycardia, Hypotension, a dry mouth and Hyperglycaemia. Ventricular arrhythmias have also been reported. Respiratory and Haemodynamic symptoms should be treated symptomatically.

Interactions with Other Medicaments: Other CNS depressant agents (Barbiturates, Narcotics, Anaesthetics, Tranquillizers, etc.) may cause additive CNS depression if used with Xylazine. Dosages of these agents may need to be reduced. Xylazine should not be used in combination with sympathomimetic drugs such as Epinephrine as ventricular arrhythmia may follow.

The concurrent intravenous use of potentiated Sulphonamides with alpha-2 agonists has been reported to cause cardiac arrhythmias which may be fatal. Intravenous administration of Trimethoprim/Sulphonamide containing products should not be undertaken when horses have been sedated with Xylazine.

Pregnancy and Lactation:

Product should not be administered during the later stages of pregnancy (particularly in cattle) except at parturition, because Xylazine causes uterine contractions and it may induce premature labour.

Do not use in cattle receiving ovum transplants as the increased uterine tone may reduce the chance of implantation of the ovum.

Side Effects / Adverse Reactions: In general, side effects, include Bradycardia, Reversible arrhythmia and Hypotension can occur. Thermoregulation can be influenced and consequently body temperature can decrease or increase depending on the ambient temperature. Depression of respiration and / or respiratory arrest can occur.

Cattle (≥ 200 kg)

- In Cattle Xylazine may induce premature parturition, and it also reduces implantation of the ovum.
- Cattle, which have received high doses of Xylazine sometimes suffer from loose faeces for 24 hours afterwards.
- Other adverse reactions include snoring, profound salivation, ruminal atony, atony of the tongue, regurgitation, bloating, nasal stridor, hypothermia, bradycardia, increased urination and reversible prolapse of the penis.

Horses

- Horses often sweat as the effects of the sedation are wearing off.
- Severe bradycardia and reduced respiratory rate have been reported especially in horses.
- Following administration to horses, a transient rise followed by a fall in blood pressure usually occurs.
- More frequent urination has been reported
- Muscle tremors and movement in response to sharp auditory or physical stimuli are possible. Although rare, violent reactions have been reported in horses following the administration of Xylazine.
- Ataxia and reversible prolapse of the penis may occur.
- In very rare cases Xylazine may induce mild colic, as the gut motility is depressed temporarily. As a preventive measure the horse should receive no feed after sedation until the effect has faded completely.

The frequency of adverse reactions is defined using the following convention:

- Very common (more than 1 in 10 animals displaying adverse reaction(s) during the course of one treatment)
- Common (more than 1 but less than 10 animals in 100 animals)
- Uncommon (more than 1 but less than 10 animals in 1,000 animals)
- Rare (more than 1 but less than 10 animals in 10,000 animals)
- Very rare (less than 1 animal in 10,000 animals, including isolated reports).

Signs and Symptoms of Overdose and Treatment:

In the event of an accidental overdose, Cardiac arrhythmias, Hypotension, and profound CNS and respiratory depression may occur. Seizures have also been reported after an overdose. Xylazine can be antagonized by α_2 -adrenergic antagonists.

To treat the respiratory depressant effects of Xylazine, mechanically respiratory support with or without respiratory stimulants (e.g. Doxapram) can be recommended.

Withdrawal Period(s):

Cattle (≥ 200 kg):

Meat and offal: 1 day

Milk: zero hours

Horses:

Meat and offal: 1 day

Milk: zero hours

Shelf life after first opening of product: 28 days.

Storage Conditions: Do not store above 30 °C.

Dosage forms and packing available: Injection and 50 ml Amber coloured Glass vials.

Incompatibilities: In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other veterinary medicinal products.

Applicant/Marketing Authorisation Holder

M/s. AMIPRO SDN.BHD. (Co.No.1166264-V).

Level 3, Bangunan Inkubator Inovasi
Universiti (I2u), Sains@Usm, Lebu
Bukit Jambul, 11900 Bayan Lepas
Pulau Pinang, Malaysia

Manufacturing site Address

Nectar Laboratories Private Limited

Works at: Plot No: 54 A, 54 B & 55, Anrich
Industrial Estate, Bollaram Village, Jinnaram
Mandal, Medak Dist., Telangana state - INDIA

Last revision date: February 14th, 2023